

1. A.M. - G.M. Inequality (算術及幾何平均不等式).

For any n ($n > 1$) non-negative real numbers $a_1, a_2, a_3, \dots, a_n$,

$$\frac{a_1 + a_2 + a_3 + \dots + a_n}{n} \geq (a_1 a_2 a_3 \dots a_n)^{\frac{1}{n}}, \quad (\text{or } a_1^n + a_2^n + a_3^n + \dots + a_n^n \geq n a_1 a_2 a_3 \dots a_n)$$

The equality holds if and only if $a_1 = a_2 = a_3 = \dots = a_n$.

Proof: (Method 1) Step 1 (Mathematical Induction on $n=2^m$, for $m = 1, 2, 3, \dots$)

When $n=2^1$, Since $(\sqrt{a_1} - \sqrt{a_2})^2 \geq 0$, hence $a_1 + a_2 \geq 2\sqrt{a_1 a_2}$

That is $\frac{a_1 + a_2}{2} \geq \sqrt{a_1 a_2}$. Hence the proposition is true when $m=1$.

Next, assume that the proposition is true when $n = 2^k = p$, for $k = 1, 2, 3, \dots$

That is $\frac{a_1 + a_2 + a_3 + \dots + a_p}{p} \geq (a_1 a_2 a_3 \dots a_p)^{\frac{1}{p}}$

When $n = 2^{k+1} = 2(2^k) = 2p$, consider any $2^{k+1} (= 2p)$ non-negative real numbers $b_1, b_2, b_3, \dots, b_{2p}$.

Let $a_1 = \frac{b_1 + b_2}{2}, a_2 = \frac{b_3 + b_4}{2}, \dots, a_p = \frac{b_{2p-1} + b_{2p}}{2}$.

$$\begin{aligned} \text{Based on our assumption, } & \frac{b_1 + b_2 + b_3 + \dots + b_{2p}}{2p} \\ &= \left(\frac{b_1 + b_2}{2} + \frac{b_3 + b_4}{2} + \dots + \frac{b_{2p-1} + b_{2p}}{2} \right) \left(\frac{1}{p} \right) \\ &\geq \left[\left(\frac{b_1 + b_2}{2} \right) \left(\frac{b_3 + b_4}{2} \right) \dots \left(\frac{b_{2p-1} + b_{2p}}{2} \right) \right]^{\frac{1}{p}} \\ &\geq \left[\sqrt{b_1 b_2} \sqrt{b_3 b_4} \dots \sqrt{b_{2p-1} b_{2p}} \right]^{\frac{1}{p}} \\ &= (b_1 b_2 b_3 b_4 \dots b_{2p-1} b_{2p})^{\frac{1}{2p}} \end{aligned}$$

Hence the proposition is true for $n = 2^m$, for any positive integer m .

Step 2: Given any n ($n \neq 2^m$) non-negative real numbers $a_1 + a_2 + a_3 + \dots + a_n$, choose m

such that $2^{m-1} < n < 2^m$, and add $(2^m - n)$ times the number $A = \frac{a_1 + a_2 + a_3 + \dots + a_n}{n}$

$$\begin{aligned} \text{so that } & \frac{a_1 + a_2 + \dots + a_n + (2^m - n)A}{2^m} \geq \sqrt[2^m]{a_1 a_2 \dots a_n A^{(2^m - n)}} \\ & A \geq \sqrt[2^m]{a_1 a_2 \dots a_n A^{(2^m - n)}} \\ & A^{1 - \left(\frac{2^m - n}{2^m}\right)} \geq \sqrt[2^m]{a_1 a_2 \dots a_n} \\ & A^{\frac{n}{2^m}} \geq (a_1 a_2 \dots a_n)^{\frac{1}{2^m}} \\ & A \geq (a_1 a_2 a_3 \dots a_n)^{\frac{1}{n}} \end{aligned}$$

(Method 2):

For any n ($n > 1$) non-negative real numbers $a_1, a_2, a_3, \dots, a_n$, let G be the geometric mean.

That is $G = (a_1 a_2 a_3 \dots a_n)^{\frac{1}{n}}$

If a_1 and a_2 are respectively the smallest and largest numbers in the set, obviously

$a_1 < G < a_2$. Now let $x = \frac{a_1 a_2}{G}$, again obviously $a_1 < x < a_2$.

Now consider $(a_1 + a_2) - (G + x) = (a_1 + a_2) - (G + \frac{a_1 a_2}{G})$

$$= [G(a_2 - G) + a_1 (G - a_2)] \left(\frac{1}{G} \right) = [(G - a_1)(a_2 - G)] \left(\frac{1}{G} \right) > 0$$

Hence $(a_1 + a_2) > (G + x)$, and $\frac{a_1 + a_2 + a_3 + \dots + a_n}{n} \geq \frac{G + x + a_3 + \dots + a_n}{n}$

Note that the geometric mean is unchanged.

Next, we leave the G alone and repeat the procedure on the remaining $n-1$ numbers, we get

$$\frac{G + x + a_3 + \dots + a_n}{n} \geq \frac{G + G + y + \dots + a_n}{n}$$

The procedure can be repeated until we change all the numbers to G .

Hence $\frac{a_1 + a_2 + a_3 + \dots + a_n}{n} \geq \frac{G + G + G + \dots + G}{n} = G = (a_1 a_2 a_3 \dots a_n)^{\frac{1}{n}}$

Method 3 (Using Mathematical Induction)

For $n = 2$, $(\sqrt{a} - \sqrt{b})^2 \geq 0$, hence $\frac{a+b}{2} \geq \sqrt{ab}$.

Assume that the proposition is true for any $n-1$ non-negative real numbers.

That is $\frac{a_1 + a_2 + \dots + a_{n-1}}{n-1} \geq (a_1 a_2 \dots a_{n-1})^{\frac{1}{n-1}}$.

Consider any n non-negative real numbers, $a_1, a_2, \dots, a_n$.

Without loss of generality, assume that $a_1 \leq a_2 \leq \dots \leq a_n$.

Further, if $a_1 = a_2 = \dots = a_n$, the inequality holds, otherwise $\frac{a_1 + a_2 + \dots + a_{n-1}}{n-1} < a_n$,

$$\begin{aligned} \text{Now } \frac{1}{n}(a_1 + a_2 + \dots + a_{n-1} + a_n) &= \frac{1}{n}[(n-1) \left(\frac{a_1 + a_2 + \dots + a_{n-1}}{n-1} \right) + a_n] \\ &= \frac{a_1 + a_2 + \dots + a_{n-1}}{n-1} + \frac{a_n - \frac{a_1 + a_2 + \dots + a_{n-1}}{n-1}}{n} \end{aligned}$$

Using the inequality $(x+y)^n \geq x^n + nx^{n-1}y$, we have

$$\begin{aligned} \left[\frac{1}{n}(a_1 + a_2 + \dots + a_{n-1} + a_n) \right]^n &\geq \left(\frac{a_1 + a_2 + \dots + a_{n-1}}{n-1} \right)^n + n \left(\frac{a_1 + a_2 + \dots + a_{n-1}}{n-1} \right)^{n-1} \left(\frac{a_n - \frac{a_1 + a_2 + \dots + a_{n-1}}{n-1}}{n} \right) \\ &= a_n \left(\frac{a_1 + a_2 + \dots + a_{n-1}}{n-1} \right)^{n-1} \\ &\geq a_n (a_1 a_2 \dots a_{n-1}) \\ &= a_1 a_2 \dots a_{n-1} a_n \end{aligned}$$

Hence $\frac{1}{n}(a_1 + a_2 + \dots + a_{n-1} + a_n) \geq (a_1 a_2 \dots a_{n-1} a_n)^{\frac{1}{n}}$.

The proposition is also true when we have n non-negative real numbers..

By the Principle of Mathematical Induction, the proposition is true for any positive integer n .

Ex. 1: Let x, y and z be positive real numbers such that $x + y + z = 1$. Prove that

$$(i) x^2 y z \leq \frac{1}{64}, \quad (ii) \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \geq 9, \quad (iii) \left(\frac{1}{x} - 1 \right) \left(\frac{1}{y} - 1 \right) \left(\frac{1}{z} - 1 \right) \geq 8.$$

Ex. 2: Prove that if a and b are positive real numbers such that $a+b=1$, then $(a + \frac{1}{a})(b + \frac{1}{b}) \geq \frac{25}{4}$.

Ex. 3: For any positive real numbers a, b and c , prove that $\frac{c}{a} + \frac{a}{b+c} + \frac{b}{c} \geq 2$. When does the equality hold?

Ex. 4: Prove that if a, b, c are positive real numbers such that $a+b+c = 1$, then $a^3 + b^3 + c^3 \geq \frac{1}{9}$.

Ex. 5: Given that $a > b > c > 0$, find the minimum of $2a^2 + \frac{1}{ab} + \frac{1}{a(a-b)} - 6ac + 9c^2$.

When does this happen?

Ex. 6: For real numbers $a > 1, b > 1$, prove that $\frac{a^2}{b-1} + \frac{b^2}{a-1} \geq 8$.

Ex. 7: Let a, b be real numbers such that $a+b = 17$. Find the smallest value of $2^a + 4^b$.

Ex. 8: For positive values of x , find the minimum value of $x^3 - 3x + 5$.

Ex. 9: For any positive integer $n > 1$, prove that

$$(a) \ 2.4.6 \dots (2n) < (n+1)^n. \quad (b) \ (n!)^3 < n^n \left(\frac{n+1}{2}\right)^{2n}$$

Ex. 10: Prove that $37^{73} > 73!$.

Ex. 11: Given that $\cos^2 A + \cos^2 B + \cos^2 C = 1$, prove that $\cot^2 A + \cot^2 B + \cot^2 C \geq \frac{3}{2}$.

Ex. 12: Given that $a_1, a_2, \dots, a_n$ are n positive real numbers such that $a_1 a_2 \dots a_n = 1$.
Prove that $(2+a_1)(2+a_2) \dots (2+a_n) \geq 3^n$.

Ex. 13: Let $a, b, c \in \mathbb{R}$ such that $a^2 + b^2 + c^2 = 2$. Prove that $a+b+c \leq abc + 2$.

Ex. 14: Let a, b, c and d be positive numbers, prove that

$$\frac{1}{a+b+c} + \frac{1}{b+c+d} + \frac{1}{c+d+a} + \frac{1}{d+a+b} \geq \frac{16}{3(a+b+c+d)}.$$

Ex. 15: Let a, b, c, d be positive real numbers such that $a+b+c+d = 4$. Prove that

$$\frac{a}{b^2+1} + \frac{b}{c^2+1} + \frac{c}{d^2+1} + \frac{d}{a^2+1} \geq 2.$$

Ex. 16: Prove that for any positive integer n , $1 < \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{3n+1} < 2$.

2. Weighted AM-GM Inequality (加權算術及幾何平均不等式)

Let $a_1, a_2, \dots, a_n$ be non-negative real numbers, and $\phi_1, \phi_2, \dots, \phi_n$ be non-negative real numbers such that $\phi_1 + \phi_2 + \dots + \phi_n = 1$, then

$$\phi_1 a_1 + \phi_2 a_2 + \dots + \phi_n a_n \geq (a_1^{\phi_1})(a_2^{\phi_2}) \dots (a_n^{\phi_n})$$

In other words, $\sum_{i=1}^n \phi_i a_i \geq \prod_{i=1}^n (a_i^{\phi_i})$

Proof for the case $n = 2$:

Let p and q be non-negative integers and x, y be non-negative real numbers.

Take p 'x's and q 'y's, and using A.M. $\geq$ G.M., $\frac{px+qy}{p+q} \geq (x^p y^q)^{\frac{1}{p+q}}$

$$\text{Let } \alpha = \frac{p}{p+q}, \text{ and } \beta = \frac{q}{p+q}, \text{ then } \alpha + \beta = 1$$

$$\text{The inequality becomes } \alpha x + \beta y \geq x^\alpha y^\beta.$$

The proof for the general case is similar.

Ex. 17: The real numbers a, b, c, d are such that $a \geq b \geq c \geq d > 0$ and $a+b+c+d = 1$.
Prove that $(a+2b+3c+4d) a^{ab} b^{bc} c^{cd} d^d < 1$. (62th IMO (2020) No. 2)

3. Cauchy-Schwarz Inequality (柯西不等式)

For any two sequences of non-negative real numbers $a_1, a_2, a_3, \dots, a_n$ and $b_1, b_2, b_3, \dots, b_n$,

$$[a_1^2 + a_2^2 + a_3^2 + \dots + a_n^2] [b_1^2 + b_2^2 + b_3^2 + \dots + b_n^2] \geq (a_1 b_1 + a_2 b_2 + a_3 b_3 + \dots + a_n b_n)^2$$

The equality holds if and only if $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} = \dots = \frac{a_n}{b_n}$.

$$\text{Proof (Method 1)} : \left(\sum_{i=1}^n a_i^2\right) \left(\sum_{i=1}^n b_i^2\right) - \left(\sum_{i=1}^n a_i b_i\right)^2 = \sum_{i=1}^{n-1} \sum_{j=i+1}^n (a_i^2 b_j^2 + a_j^2 b_i^2) - 2 \sum_{i=1}^{n-1} \sum_{j=i+1}^n (a_i b_i a_j b_j)$$

$$= \sum_{i=1}^{n-1} \sum_{j=i+1}^n (a_i b_j - a_j b_i)^2 \geq 0, \quad \text{Hence } \left(\sum_{i=1}^n a_i^2\right) \left(\sum_{i=1}^n b_i^2\right) \geq \left(\sum_{i=1}^n a_i b_i\right)^2$$

Equality holds only when $a_i b_j = a_j b_i$ for all i, j ; i.e. $\frac{a_i}{b_i} = \frac{a_j}{b_j}$ for all i, j .

(Method 2): Consider the inequality $\sum_{i=1}^n (a_i x - b_i)^2 \geq 0$, i.e. $(\sum_{i=1}^n a_i^2) x^2 - 2(\sum_{i=1}^n a_i b_i) x + (\sum_{i=1}^n b_i^2) \geq 0$

The necessary condition is (i) $(\sum_{i=1}^n a_i^2) > 0$ and (ii) $(2 \sum_{i=1}^n a_i b_i)^2 \leq 4(\sum_{i=1}^n a_i^2)(\sum_{i=1}^n b_i^2)$

Hence $(\sum_{i=1}^n a_i^2)(\sum_{i=1}^n b_i^2) \geq (\sum_{i=1}^n a_i b_i)^2$

Equality holds only when $a_i x - b_i = 0$ for all i ; i.e. $\frac{a_i}{b_i} = \frac{a_j}{b_j}$ for all i, j .

Ex. 18: Prove that for any sequence of n real numbers $a_1, a_2, \dots, a_n$,

$$(i) \quad \sum a_i \leq \sqrt{n \sum a_i^2}$$

(ii) If all the numbers are positive, then $(\sum a_i)(\sum \frac{1}{a_i}) \geq n^2$.

Ex. 19: Prove that for any real numbers a, b, c, d , $(a+b+c+d)^2 \leq 3(a^2 + b^2 + c^2 + d^2) + 6ab$.

Ex. 20: Let a be a real number, prove that $2\sqrt{a-4} + 3\sqrt{5-a} \leq \sqrt{13}$. When does the equality hold?

Ex. 21: Given that $5x^2 + 2y^2 = 10$, find that maximum value of $3x + 4y$.

Ex. 22: Let a, b and c be real numbers such that $a + b + c = 1$. Find the minimum value of $2a^2 + 3b^2 + c^2$. When does the equality hold?

Ex. 23: If $a > 0, b > 0$ and $a + b = 1$, find the minimum value of $(\frac{1}{a^2} - 1)(\frac{1}{b^2} - 1)$.

Ex. 24: If a, b, c, x, y, z are positive real numbers such that $a^2 + b^2 + c^2 = 36, x^2 + y^2 + z^2 = 49$, and $ax + by + cz = 42$, find the value of $\frac{a+b+c}{x+y+z}$.

Ex. 25: Given $a > -\frac{1}{2}$ and $b > -1$, and $a+b=1$, find the maximum value of $\sqrt{2a+1} + \sqrt{b+1}$.

Ex. 26: Let $a_1, a_2, \dots, a_n$ be n ($n > 1$) positive real numbers, and m be their arithmetic mean. Prove that $nm^2 \leq a_1^2 + a_2^2 + \dots + a_n^2 < n^2 m^2$

Ex. 27: Prove that if a, b, c, d are positive real numbers, then $(a+b+c+d)(a^3+b^3+c^3+d^3) \geq (a^2+b^2+c^2+d^2)^2$.

Ex. 28: Prove that for any positive integer n , $1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} < \sqrt{2n-1}$

Ex. 29: Let n be a positive integer not less than 2. Prove that

$$\frac{4}{7} < 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{2n-1} - \frac{1}{2n} < \frac{\sqrt{2}}{2}.$$

Ex. 30: Given that a, b, c, d and e are real numbers such that $a + b + c + d + e = 8$, and $a^2 + b^2 + c^2 + d^2 + e^2 = 16$. Find the maximum and minimum value of e .

Ex. 31: If a, b, c are the lengths of the three sides of a triangle, prove that

$$\frac{a^2}{b+c-a} + \frac{b^2}{c+a-b} + \frac{c^2}{a+b-c} \geq a + b + c.$$

Ex. 32: Let a, b and c be positive real numbers such that $abc = 1$. Prove that

$$\frac{1}{a^3(b+c)} + \frac{1}{b^3(c+a)} + \frac{1}{c^3(a+b)} \geq \frac{3}{2}. \quad (36^{\text{th}} \text{ IMO (1995) No. 2})$$

4. Harmonic Mean & Root Mean Square (調和平均與均方根)

(a) For any n ($n > 1$) positive real numbers $a_1, a_2, a_3, \dots, a_n$, the Harmonic Mean (調和平均)

$$H_n = \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \dots + \frac{1}{a_n}} \text{ and the Root Mean Square (均方根) is R.M.S.} = \sqrt{\frac{a_1^2 + a_2^2 + \dots + a_n^2}{n}}$$

Theorem : R.M.S. $\geq$ A.M. $\geq$ G.M $\geq$ H.M.

(i) **To prove that H.M. $\leq$ G.M.:**

$$\text{Since } \left(\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \dots + \frac{1}{a_n}\right) \left(\frac{1}{n}\right) \geq \left(\frac{1}{a_1 a_2 a_3 \dots a_n}\right)^{\frac{1}{n}}$$

$$\text{Hence } \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \dots + \frac{1}{a_n}} \leq (a_1 a_2 a_3 \dots a_n)^{\frac{1}{n}}$$

(ii) **To proof that R.M.S. $\geq$ A.M.:**

$$\text{Using Cauchy's Inequality, } (a_1^2 + a_2^2 + \dots + a_n^2)(1+1+\dots+1) \geq (a_1 + a_2 + \dots + a_n)^2$$

$$\text{Hence } n(a_1^2 + a_2^2 + \dots + a_n^2) \geq (a_1 + a_2 + \dots + a_n)^2$$

$$\text{It means that } \sqrt{\frac{a_1^2 + a_2^2 + \dots + a_n^2}{n}} \geq \frac{a_1 + a_2 + a_3 + \dots + a_n}{n}$$

Alternative Method:

$$\begin{aligned} & \left(\frac{a_1^2 + a_2^2 + \dots + a_n^2}{n}\right) - \left(\frac{a_1 + a_2 + a_3 + \dots + a_n}{n}\right)^2 = \frac{1}{n^2} [n(a_1^2 + a_2^2 + \dots + a_n^2) - (a_1 + a_2 + \dots + a_n)^2] \\ & = \frac{1}{n^2} [(n-1)(a_1^2 + a_2^2 + \dots + a_n^2) - 2 \sum_{i=1}^{n-1} \sum_{j=i+1}^n a_i a_j] = \frac{1}{n^2} \sum_{i=1}^{n-1} \sum_{j=i+1}^n (a_i - a_j)^2 \geq 0 \end{aligned}$$

$$\text{Hence } \sqrt{\frac{a_1^2 + a_2^2 + \dots + a_n^2}{n}} \geq \frac{a_1 + a_2 + a_3 + \dots + a_n}{n}$$

Equality holds if and only if $a_i = a_j$ for $1 \leq i, j \leq n$.

Ex. 33: Let a and b be positive real numbers such that $a + b = 1$. Prove that

$$(a) \sqrt{2a+1} + \sqrt{2b+1} \leq 2\sqrt{2}; \quad (b) a^4 + b^4 \geq 0.125.$$

Ex. 34: Let a, b, c be positive real numbers satisfying $a^2 + b^2 + c^2 = 3$. Prove that

$$\frac{1}{1+2ab} + \frac{1}{1+2bc} + \frac{1}{1+2ca} \geq 1$$

Ex. 35: If $a_1 + a_2 + a_3 + \dots + a_n = 1 + \frac{1}{2n}$, prove that

$$\sqrt{2a_1+1} + \sqrt{2a_2+1} + \dots + \sqrt{2a_n+1} \leq n+1$$

Ex. 36: Given that x, y are positive real numbers and $x+y < 27$, prove that $\frac{\sqrt{x}+\sqrt{y}}{\sqrt{xy}} + \frac{1}{\sqrt{27-x-y}} \geq 1$.

Ex. 37: For any positive real numbers x, y and z , prove that

$$\frac{xyz(x+y+z+\sqrt{x^2+y^2+z^2})}{(x^2+y^2+z^2)(xy+yz+zx)} \leq \frac{3+\sqrt{3}}{9}.$$

Ex. 38. For any two sequences of non-negative real numbers $a_1, a_2, a_3, \dots, a_n$ and $b_1, b_2, b_3, \dots, b_n$, prove

$$\text{that } \sqrt{a_1^2 + a_2^2 + \dots + a_n^2} + \sqrt{b_1^2 + b_2^2 + \dots + b_n^2} \geq \sqrt{(a_1 + b_1)^2 + (a_2 + b_2)^2 + \dots + (a_n + b_n)^2}$$

When does the inequality hold? (This is called Triangle Inequality (三角不等式))

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